

Assignment:- 5

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Subject: Software Engineering

① Analyze Component Based Software Engineering (CBSE)

→ Component based Software Engineering is an approach where software systems are built using reusable, pre-existing components.

Key points:

- Focuses on reusability and modularity.
- Components are independent, reusable units with well-defined interfaces.
- Reduces development time and cost.

Advantages

- Faster development due to reuse.
- Improved reliability (tested components).
- Easy maintenance and scalability.

Disadvantages

- Integration issue between components.
- Dependency on third-party components
- version compatibility problems.

② Discuss web Engineering lifecycle.

→ web Engineering deals with systematic development of web applications.

Phases:

- 1) Requirement Analysis
 - Identify user needs and goals.
- 2) Design
 - UI/UX design, navigation structure, database design
- 3) Development
 - Frontend & Backend coding.
- 4) Testing
 - Functional, usability, performance testing.
- 5) deployment
 - Hosting the application on servers

6) Maintenance:

- updates, bug fixes, content changes.

Key Feature:

- Highly interactive and user-driven.

③ Explain Reverse Engineering process.

→ Reverse Engineering is the process of analyzing existing software to understand its design and functionality.

Steps:

- Information Gathering - collect code, documents.
- Code Analysis - study structure and logic.
- Abstraction - Identify higher-level design.
- Documentation - Recreate system design documents.

Uses:

- System maintenance
- Understanding legacy systems
- Security analysis

④ Analyze the challenges of Big data in Software Engineering.

⇒ Big Data introduces complexity due to massive data size and variety.

Major challenges:

- Data volume - Handling huge datasets.
- Data variety - structured, semi-structured, unstructured data.
- Data velocity - Real-time data processing.
- Storage issues - Efficient storage management.
- Security & Privacy - protecting sensitive data.
- Scalability - systems must scale dynamically.

⑤ Discuss mobile Application Development Challenges.

→ Developing mobile apps involves multiple constraints.

Challenges:

- Device Fragmentation: Different screen sizes, OS versions.
- Performance Optimization: Limited battery & memory.
- Network Dependency: Handling poor connectivity.
- Security Risk - Data leakage, malware
- User Experience - Need for intuitive UI
- Cross platform issues - Android vs iOS differences.

⑥ Explain Integration of CASE Tools in SDLC.

→ CASE (Computer - Aided Software Engineering) tools supports SDLC activities.

Integration Across Phases:

- Planning - Project management tools.
- Analysis & Design → UML modeling tools
- Development - code generating tools
- Testing - Automated testing tools
- Maintenance - version control and debugging tools.

Benefits:

- Improves productivity
- Reduces human errors
- Ensures consistency and documentation

⑦ Analyze future trends in software Engineering Technologies.

→ Software engineering is evolving rapidly with new technologies.

Key Trends:

- AI & ML: Automated coding, testing.
- Dev ops & CI/CD - faster delivery cycles
- Cloud computing - Scalable and distributed systems.
- Microservices Architecture - modular and flexible systems
- Blockchain - secure and transparent systems.
- Low code / No code platform: faster development with minimal coding.

Impact:

- Increased automation
- Faster development cycles
- Higher demand for skilled engineers.

Shruti
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